

Highlights

Transport-Inspired Rare-Event Sampling in Spin Systems: Learned Importance Coordinates Improve Access to Deep Targets Across Independent Disorder Realizations

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- Adjoint importance and the rare-event committor share the same backward-conditioning structure for first-passage problems, and coincide for an absorbing target.
- Exact tests validate both the zero-variance importance limit and unbiased WE population control.
- At matched update cost and full replication ($R = 20$), WE holds an approximately two-fold precision advantage over naive sampling at the deepest resolved Ising-tail bin, though not uniformly across the tail.
- Learned importance improves deep-event discovery across six independent Edwards–Anderson disorder realizations (67.2% versus 52.8%, $p = 0.0071$).
- The reliability improvement does not yet translate into wall-clock efficiency (0.33, 95% CI [0.19, 0.58]).

Transport-Inspired Rare-Event Sampling in Spin Systems: Learned Importance Coordinates Improve Access to Deep Targets Across Independent Disorder Realizations

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ABSTRACT

Rare-event sampling requires computational effort to be concentrated in configurations that disproportionately determine a target probability. Radiation transport addresses this through adjoint importance functions and weight-window population control, and this work transfers that toolkit to rare-event sampling in lattice spin systems, using the shared backward-conditioning structure of the transport adjoint and the transition-path committor. A solvable birth–death process reproduces the zero-variance limit of exact importance, and enumeration of a 4×4 lattice confirms that weight-conserving Weighted Ensemble (WE) resampling is unbiased for any binning coordinate. At matched spin-update cost and full replication ($R = 20$), WE and naive sampling resolve the $L = 20$ Ising magnetization tail to comparable depth, with WE holding a modest, roughly two-fold precision advantage at the single deepest bin either method reaches. For the frustrated Edwards–Anderson spin glass, a self-consistency-trained neural coordinate acting on the full spin configuration bins the weighted population. Across six independent disorder realizations with one trained network each, the deepest calibrated target is reached in 121 of 180 replicas (67.2%) against 95/180 (52.8%) for the energy-binned coordinate (Fisher exact $p = 0.0071$); the same advantage holds within a single realization, across three independently trained networks compared against a matched-size baseline (66.7% versus 48.9%, $p = 0.023$). That improvement is one of reliability rather than wall-clock efficiency. Repeated neural inference costs four to five times the wall-clock time of the energy coordinate, so the directly measured figure of merit sits at roughly one-third of parity (0.33, 95% CI [0.19, 0.58]). The transport toolkit transfers, then, and the reliability advantage generalizes across independent disorder realizations rather than being tied to one lattice.

1. Introduction

A rare event is one that direct simulation almost never observes: the nucleation of a new phase, a protein crossing between conformational basins [1], or a detector registering a particle that has penetrated many mean free paths of shielding [2, 3]. In each case the probability of the event is itself the quantity of scientific interest, and it is exactly that quantity which unbiased dynamics represents worst. If an event has equilibrium or first-passage probability $p \ll 1$, the relative error of a direct Bernoulli estimator formed from N effectively independent samples follows the elementary result [2, 3]

$$\epsilon_{\text{rel}} = \sqrt{\frac{1-p}{Np}} \simeq (Np)^{-1/2}, \quad p \ll 1, \quad (1)$$

so the sample count required for fixed relative precision grows as $1/p$ and each additional decade of rarity costs an order of magnitude in computation. The failure is structural rather than incidental, and every established remedy responds to it the same way: not with more samples, but with better-placed ones. This elementary scaling is the origin of a broad family of rare-event techniques, including importance sampling, splitting, transition-path sampling, Forward Flux Sampling (FFS), Adaptive Multilevel Splitting (AMS), and Weighted Ensemble (WE) methods [1–3, 16, 20, 22, 38]. What each of them requires in practice is a rule for deciding which samples are the better-placed ones.

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Two mature traditions answer that question using closely related mathematical objects. In transition-path theory, the committor $h(s)$ is the probability that a trajectory initiated at configuration s reaches a target set before a competing boundary, and it provides an optimal reaction coordinate for that event [4–6]. In classical particle transport, an adjoint solution assigns an importance to a phase-space point according to its expected contribution to a detector response; this importance can be used to bias sources, alter transition probabilities, and construct weight windows [7–11]. The formal connection to the Doob h -transform and conditioned Markov processes is well known [12–14]. The contribution of the present work is not to claim a new identity between these objects, but to carry out a systematic transfer and validation of the transport variance-reduction viewpoint in spin-system rare-event sampling, together with a controlled test of a learned configuration-space coordinate built on this viewpoint.

The transport perspective is attractive because it separates the ideal importance function from the population-control mechanism used when that importance is only approximate. Exact adjoint biasing can in principle produce zero-variance estimators, while weight windows use splitting and roulette to keep statistical weight near a prescribed range. This distinction underlies CADIS and related hybrid deterministic–Monte Carlo variance-reduction methods [10, 11, 15]. In the present setting, the same logic motivates using a committor-like quantity to organize a weighted population while retaining the original spin dynamics.

Rare-event population methods have a parallel history outside transport. WE was introduced for biomolecular transition sampling by Huber and Kim, shown to be statistically exact for broad classes of dynamics and binning procedures, and made practical for complex molecular systems by steady-state formulations and mature software [16–19]; FFS and AMS address related first-passage problems through interfaces or adaptive levels [20–22], while transition-path sampling works with reactive-path ensembles and can avoid prescribing a one-dimensional coordinate at all [1, 23]. Machine learning meanwhile offers several routes to the high-dimensional objects these methods need: flow-based and autoregressive samplers for equilibrium distributions [24–27], neural approximations of committor and Feynman–Kac observables from short trajectories [28, 29], learned bias potentials and importance functions [30, 31], and neural Doob dynamics for large deviations [32]. Together these motivate the central practical question of this paper: when the rare event occurs in a high-dimensional spin configuration space, how much is gained by replacing a hand-designed coordinate with a learned, committor-inspired one, measured against a strong physical baseline and with inference cost charged to the result?

The study is organized as a validation ladder rather than a single benchmark, built around three questions. Is the transport/committor correspondence mathematically valid in the formulated first-passage setting? Does weight-conserving WE genuinely extend accessible rare-event tails at matched cost? And does a learned full-configuration coordinate improve deep-event access over a strong physical energy coordinate, reproducibly across independent instances of the problem rather than for one particular realization? A solvable one-dimensional birth–death chain first verifies the zero-variance limit of exact importance. Exact enumeration of a 4×4 Ising lattice then verifies unbiased WE resampling. A larger $L = 20$ Ising calculation tests whether WE genuinely extends a rare magnetization tail at matched spin-update cost, and finally the Edwards–Anderson (EA) spin glass provides a frustrated landscape in which magnetization is a poor coordinate and the low-energy target is a more demanding test for learned configuration-space importance [33–36]. Throughout, *reliability* (whether a method reaches the event at all, reproducibly) is distinguished from *efficiency* (variance reduction per unit of actual computational cost). That distinction matters most for neural coordinates, since inference overhead is not represented by a spin-update count.

The main findings are positive but bounded. WE is unbiased, validated against exact references, and holds a modest precision advantage over naive sampling at the deepest resolved point of the Ising tail, though not uniformly across it. In the EA model, a self-consistency-trained neural coordinate significantly improves deep-event discovery over the physical energy coordinate, an advantage reproduced both across independently trained networks on one disorder realization and across six independent disorder realizations. That improvement is a reliability result, though: once inference is included in the cost, the same coordinate does not improve the figure of merit. Together, these results identify a transferable theoretical core, a validated route for using learned importance information inside a weight-conserving rare-event sampler, and the specific barrier a practical implementation must still clear: inference cost, not coordinate quality.

2. Theory and Methods

2.1. Adjoint importance, committors, and zero-variance sampling

Let $K(s, s')$ be the transition kernel of a Markov chain and let B denote a target set. For a first-passage event, the forward committor is the probability of reaching B before the competing set A [4, 5]:

$$h(s) = \Pr_s(\tau_B < \tau_A). \quad (2)$$

For a discrete Markov process, this committor satisfies the backward (harmonic) equation in the interior of the state space, a standard result of transition-path theory for Markov jump processes [4–6]:

$$h(s) = \sum_{s'} K(s, s') h(s'), \quad s \notin A \cup B, \quad (3)$$

with boundary values $h = 0$ on A and $h = 1$ on B . Conditioning the dynamics with a positive harmonic function leads to the Doob h -transform, introduced in classical potential theory [12] and used in this form throughout the large-deviation and conditioned-process literature [13, 14]:

$$K_h(s, s') = K(s, s') \frac{h(s')}{h(s)}, \quad (4)$$

where defined. This is the Markov-chain counterpart of adjoint-based importance biasing in transport. With the exact h , successful paths acquire identical total statistical weight, producing the familiar zero-variance ideal.

The relationship between the two importance functions should be stated precisely, since it is weaker than an identity. The committor of Eq. (2) is a probability: the chance of reaching B before A from s , bounded in $[0, 1]$ and defined only once a competing boundary is specified. The transport adjoint, by contrast, is an expected contribution to a specified detector response; it carries the units of that response and is unbounded in general. What the two share is structure rather than definition. Both are solutions of a backward equation generated by the same transition operator, both express the value of a state with respect to a terminal objective, and both play the same role when used to tilt or to bin. They coincide as functions in the particular case of an absorbing target with unit detector cross-section and no multiplication, which is the case constructed here, but they are not universally the same object and this paper does not claim they are. What transfers is the backward-conditioning structure and the machinery built on it, not an equivalence of the two quantities in general.

This identity motivates two distinct uses of approximate importance. The first changes transition probabilities directly, which can be fragile when h is inaccurate. The second retains the original dynamics but uses population splitting and merging to control statistical weight. The second strategy is adopted here because weight conservation preserves unbiasedness even when the binning coordinate is imperfect.

2.2. Zero-variance path weighting

The zero-variance statement can be seen directly at the path level. Consider a trajectory $\omega = (s_0, s_1, \dots, s_\tau)$ generated under the Doob-transformed kernel until it reaches either A or B . Substituting the Doob transform of Eq. (4) into the path likelihood ratio gives the telescoping identity below, derived here for the discrete first-passage setting from Eqs. (3)–(4); it is the discrete-state counterpart of the classical zero-variance principle of transport importance sampling [7, 8, 11] and of the conditioned-process construction of Doob [12] and Chetrite and Touchette [14]:

$$\prod_{t=0}^{\tau-1} \frac{K(s_t, s_{t+1})}{K_h(s_t, s_{t+1})} = \prod_{t=0}^{\tau-1} \frac{h(s_t)}{h(s_{t+1})} = \frac{h(s_0)}{h(s_\tau)}. \quad (5)$$

On a successful path ending in B , $h(s_\tau) = 1$, so every successful history contributes exactly $h(s_0)$.

Fix an initial state s_0 with $h(s_0) > 0$ and let the target quantity be the first-passage probability $\pi = h(s_0) = \Pr_{s_0}(\tau_B < \tau_A)$. Generate M independent trajectories under K_h and assign each the likelihood-ratio weight of Eq. (5), giving

$$\hat{\pi} = \frac{1}{M} \sum_{m=1}^M \left[\prod_{t=0}^{\tau_m-1} \frac{K(s_t^{(m)}, s_{t+1}^{(m)})}{K_h(s_t^{(m)}, s_{t+1}^{(m)})} \right] \mathbf{1}_B(s_{\tau_m}^{(m)}). \quad (6)$$

Here $\mathbf{1}_B(\cdot)$ denotes the indicator function of the target set B , equal to 1 if its argument lies in B and 0 otherwise. States with $h = 0$ receive zero transition probability, so the transformed chain never enters A and every trajectory terminates in B ; on each such trajectory the weight telescopes to $h(s_0)$. Every term of the sum therefore equals $h(s_0)$, so $\hat{\pi} = \pi$ identically and $\text{Var}(\hat{\pi}) = 0$ for any $M \geq 1$. This holds for the exact h ; it degrades continuously, and in general non-uniformly, when h is replaced by an approximation. This is the property verified numerically in §3.1.

For an approximate importance function $\tilde{h}$, the exact telescoping property is no longer available. The present implementation separates importance representation from dynamical biasing instead: the physical Metropolis kernel is retained, while the learned or physical progress coordinate is used only to organize population splitting and merging. This design lets approximate importance information guide computational effort while weight conservation preserves the target measure.

Table 1 summarizes the operational correspondence used throughout the paper. The mapping is not intended to imply that all transport constructs carry over unchanged; source biasing, for example, has no literal analogue when the equilibrium spin distribution is fixed and the goal is to sample a stationary tail. The useful transfer is structural: an adjoint-like scalar identifies configurations that matter to the response, and a weight-management mechanism concentrates trajectory population in those configurations.

Radiation-transport concept	Markov/spin counterpart	Role in this study
Adjoint importance	Committer / event-conditioned value function	Defines ideal progress toward a rare target
CADIS importance bias	Doob h -transform / tilted dynamics	Verified in the solvable zero-variance demonstration
Weight windows	Weighted splitting and merging	Implemented through WE resampling with conserved statistical weight
Detector response	Rare-event indicator or threshold probability	Magnetization-tail or low-energy probability

Table 1: Transport concepts and their rare-event spin-system counterparts used in this work.

2.3. Spin models and rare-event targets

For the nearest-neighbor ferromagnetic Ising model, the Hamiltonian is that introduced by Ising [33] and solved exactly in two dimensions by Onsager [34],

$$H(\mathbf{s}) = -J \sum_{\langle ij \rangle} s_i s_j, \quad s_i \in \{-1, +1\}, \quad (7)$$

with periodic boundaries and $J = 1$. The principal rare observable is the absolute magnetization $|m|$. For the spin glass, the corresponding random-bond Hamiltonian is that of Edwards and Anderson [35], in the $\pm J$ form reviewed by Binder and Young [36],

$$H(\mathbf{s}) = - \sum_{\langle ij \rangle} J_{ij} s_i s_j, \quad J_{ij} \in \{-1, +1\}, \quad (8)$$

with one fixed disorder realization per comparison (three independent realizations are compared in §5.2). Quenched disorder and frustration make the low-energy states disordered, so $m \approx 0$ irrespective of energy and magnetization is not an adequate progress coordinate; low energy is used as the rare-event target instead. All reported production comparisons use single-spin Metropolis dynamics [37] at $T = 2.6$ unless stated otherwise.

2.4. Weighted Ensemble estimator and resampling

WE propagates a weighted population under the original, unbiased Metropolis dynamics for a resampling interval τ . After propagation, walkers are partitioned into bins defined by a coordinate $q(s)$. If the total statistical weight in bin b is W_b , resampling redistributes that same W_b among the target number of walkers in the bin. Splitting creates additional trajectories with reduced individual weights this way, while merging removes redundant trajectories without changing total weight. Because the dynamics themselves are not altered and the resampling operation conserves weight, the coordinate controls efficiency rather than correctness: it changes where sampling effort is placed, not the target distribution.

For a target set B , WE estimates the target probability from the total statistical weight carried by walkers in B . The estimator takes the weight-conserving form established for WE by Huber and Kim [16], proved statistically exact for broad classes of dynamics and binning procedures by Zhang et al. [17], and used in the steady-state and review formulations of Bhatt et al. [18] and Zuckerman and Chong [38]:

$$\hat{\pi}_t = \frac{\sum_i w_i^{(t)} \mathbf{1}_B(s_i^{(t)})}{\sum_i w_i^{(t)}}, \quad (9)$$

Here i indexes the walkers in the population at resampling cycle t , $w_i^{(t)}$ and $s_i^{(t)}$ are the statistical weight and state of walker i , and $\mathbf{1}_B(\cdot)$ is the same indicator function of the target set B used in Eq. (6). The reported estimate is averaged over post-burn-in resampling cycles. Three coordinates are compared: absolute magnetization $|m|$, energy per spin E/N , and a learned scalar coordinate $I_\theta(s)$. Magnetization is deliberately retained as a mismatched baseline for the EA spin glass, whereas energy is a natural physical baseline for the low-energy target.

A fixed rare-event threshold can be either trivial or physically unreachable when lattice size, temperature, or disorder changes. The computational protocol uses a short pilot simulation instead, to locate the most extreme value reached under ordinary dynamics, and then places the production threshold beyond that pilot extreme by a small number of discrete lattice steps (spacing $2/L^2$ for magnetization, $4/L^2$ for EA energy per spin). The integer “beyond” records how many such steps the target is displaced past the pilot extreme; it provides an operational depth scale, without claiming that equal values of “beyond” correspond to identical probabilities across different system sizes or disorder realizations. The resampling interval is treated as a physical parameter rather than a bookkeeping choice. Independent diagnostics in the EA model show rapid relaxation of deep configurations, so shorter resampling intervals penetrate the deep ladder more effectively at comparable cost; the production deep-rare calculation uses $\tau = 2$ for this reason.

2.5. Neural importance coordinate

The learned coordinate is a compact periodic convolutional network acting on the full spin configuration: two circularly padded 3×3 convolutional layers with nonlinear activations, followed by global mean and maximum pooling and a scalar output. This architecture is intentionally small: the research question is whether a configuration-space coordinate can discover useful progress information absent from a simple scalar observable, not whether a large neural model can fit the training set.

The training objective is shaped by the harmonic self-consistency condition of Eq. (3) itself, in the spirit of the recent label-free neural importance-sampling formulation of Kim and Cai [31], in which a learned importance function is coupled to biased transitions and branching. This direction also sits within a broader neural-committor literature: direct variational approximation of high-dimensional committors, semigroup formulations avoiding explicit second derivatives, active importance sampling for rare-event-dominated objectives, short-trajectory Feynman–Kac training, and self-consistent committor-enhanced sampling [28, 29, 39–42]. Two practical points follow from this choice of objective. First, the self-consistency condition together with the target boundary alone is under-determined, since the constant solution satisfies it everywhere; a second boundary condition anchoring bulk-equilibrium states to a distinct reference value is necessary rather than optional. §5.2 reports a case where this condition fails to be satisfiable for a particular disorder realization. Second, the present study uses the learned function more conservatively than Kim and Cai do, as a WE binning coordinate rather than inside a likelihood-ratio tilt, so that coordinate quality can be evaluated without introducing directional-tilt bias.

The trained network I_θ is accordingly termed a *learned importance coordinate*, not a neural committor: its objective is shaped by the harmonic condition of Eq. (3), but it is not claimed to be a quantitatively accurate estimate of h , and no such accuracy is measured here. Because I_θ enters only through the binning coordinate, and weight-conserving resampling is unbiased for any binning, the results below do not depend on how close I_θ is to h : an inaccurate coordinate costs efficiency, never correctness.

2.6. Computational cost, statistics, and reporting

Two cost measures are used. Matched-cost comparisons of sampling reach use the number of attempted spin updates as the common Monte Carlo work unit, natural for the Ising-tail comparison since both direct sampling and WE use the same Metropolis kernel. That measure is blind to neural inference, though, which is performed repeatedly during WE binning; wall-clock time is also recorded directly for this reason, and is the measure used whenever a figure of

merit is quoted for a neural coordinate. The figure of merit is the cost-normalized inverse variance standard in transport variance-reduction studies [8, 15],

$$\text{FOM} = \frac{1}{\sigma_{\text{rel}}^2 T}, \quad (10)$$

with T the computational cost charged to the calculation.

Suppose R independent replicas produce estimates $\hat{\pi}_r$. Following the independent-replica uncertainty reporting recommended for WE studies [38] and implemented in WESTPA [19], the across-replica relative standard deviation is

$$\sigma_{\text{rel}} = \frac{\text{sd}(\hat{\pi}_1, \dots, \hat{\pi}_R)}{\bar{\pi}}, \quad \bar{\pi} = \frac{1}{R} \sum_{r=1}^R \hat{\pi}_r, \quad (11)$$

but for deep targets the most informative quantity can instead be the fraction of independent runs that observe the event at all, so zero-hit counts are reported explicitly. Event-positive and zero-hit replicas are compared using the exact conditional test for a 2×2 contingency table introduced by Fisher [43], appropriate here because the expected counts are small. Confidence intervals on figure-of-merit ratios are obtained by the nonparametric bootstrap [44] in a two-level form that resamples the trained-network (or disorder-realization) axis in the outer loop and the replica axis in the inner loop.

3. Validation

3.1. Exact-importance limit

A biased one-dimensional birth–death chain provides a solvable test. Starting at $s_0 = 1$ with absorbing boundaries at 0 and $N = 15$ and step probability $p = 0.4$ toward the rare boundary, the exact committor is $h(s_0) = 1.144443 \times 10^{-3}$. Direct Monte Carlo gives a relative variance of 2.17×10^{-3} for 4×10^5 histories, in agreement with the Bernoulli prediction $1/(Np)$. Using the exact importance transform reduces the measured relative variance to approximately 10^{-30} , numerically reproducing the zero-variance prediction of Eq. (5).

3.2. Exact-enumeration gate for WE

For the 4×4 Ising model, all 2^{16} configurations can be enumerated. Across three validation regimes (above T_c , below T_c with the double-well structure, and the folded $|m|$ distribution), the mean $|\Delta \ln P|$ between WE and exact enumeration ranges from approximately 0.006 to 0.024; a separate learned-importance gate gives target-probability ratios of 1.000 and 1.001 relative to exact enumeration when WE is binned by magnetization and energy, respectively. These tests establish that resampling is unbiased even when the coordinate is not optimal, so later differences between coordinates are interpreted as efficiency or reliability differences, never as evidence that a coordinate changes the target measure.

Test	Reference	Outcome
Birth–death importance transform	analytic committor	exact-importance relative variance $\sim 10^{-30}$
4×4 Ising WE distribution	exact enumeration	mean $ \Delta \ln P = 0.006\text{--}0.024$ across three regimes
4×4 learned-coordinate gate	exact target probability	WE ratios 1.000 and 1.001
$L = 20$ Ising tail ($R = 20$)	matched-update naive MC	mixed: WE $\sim 2\times$ better figure of merit at the deepest resolved bin, naive $\sim 5\times$ better one bin shallower

Table 2: Validation ladder used before interpreting the larger rare-event calculations.

4. Rare-Event Reach in the Ising Ferromagnet

4.1. WE and naive sampling in the Ising tail

At $L = 20$ and matched update cost ($\sim 3.4 \times 10^8$ spin updates), naive sampling and WE agree closely through the bulk and into the near tail, with $R = 20$ independent replicas per method. At the second-deepest resolved bin

($|m| = 0.886$), naive sampling is in fact the more precise of the two at matched cost: relative standard deviation 0.36 against WE's 0.79, a figure-of-merit advantage for naive of roughly 5 $\times$. Only at the single deepest bin reached by either method ($|m| = 0.932$, $P \approx 1.1 \times 10^{-7}$ for WE against $P \approx 7.4 \times 10^{-8}$ for naive) does WE hold an edge, with a figure of merit about 2 $\times$ that of naive (relative standard deviation 3.10 versus 4.47); both estimates are individually noisy at this depth.

An earlier characterization of this comparison, using four replicas instead of twenty, reported that naive sampling returned no estimate at all beyond $|m| = 0.886$ and framed WE's reach as roughly 24 $\times$ deeper. That framing does not survive replication: the deepest bin any replica happens to reach is an extreme-value statistic that recedes as more replicas are drawn, not a stable property of naive sampling, and at $R = 20$ naive does reach the same deepest bin as WE, just far less reliably. The honest content of this comparison is accordingly narrower than first reported. WE holds a real, roughly two-fold precision advantage at the single deepest resolved point, but this is not a uniform advantage across the tail, and the comparison should not be read as WE systematically outperforming naive sampling at this system size. The primary evidence that WE is correctly implemented and unbiased remains the exact validation of §3, which this comparison does not affect.

4.2. Calibration of the deep-rare spin-glass regime

The neural-coordinate experiment is placed deliberately beyond the range where direct sampling is consistently informative. Table 3 summarizes the depth calibration at $L = 16$ and identifies the access-dominated regime used for the learned-coordinate test.

Beyond	$\hat{\pi}$	relative s.d.	zero-hit replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0	—	10/10

Table 3: Depth calibration for the EA model at $L = 16$ using 10 independent direct-sampling replicas. The integer “beyond” indexes successively deeper discrete energy thresholds.

The beyond= 4 threshold is used as a stress test of rare-event access for this reason: direct Monte Carlo is fully blind there, so the learned-coordinate comparison directly tests whether a configuration-space importance map can organize the weighted population so that trajectories continue to enter a region essentially absent from the direct Monte Carlo sample.

5. Learned Importance in the Deep-Rare Spin Glass

At beyond= 4, direct Monte Carlo is effectively blind under the calibrated budget. The WE budget was increased to 80 bins, 40 walkers per occupied bin, 3000 iterations, and $\tau = 2$.

5.1. One disorder, multiple trained networks

Three independently trained self-consistency networks were compared with the same physical-coordinate baselines on one EA disorder realization ($L = 16$, $T = 2.6$), 30 replicas per method. Using independently trained networks establishes a reproducible coordinate-level effect rather than a favorable outcome from one particular initialization.

Method	event-positive replicas	discovery rate
naive	1/30	3.3%
WE[m]	5/30	16.7%
WE[E] (balanced, $n = 90$)	44/90	48.9%
WE[I_θ], seed 0	19/30	63.3%
WE[I_θ], seed 1	20/30	66.7%
WE[I_θ], seed 2	21/30	70.0%
WE[I_θ], pooled ($n = 90$)	60/90	66.7%

Table 4: Deep-rare EA comparison at $L = 16$, beyond= 4, one disorder realization. Each learned-coordinate row is a separately trained network evaluated over 30 replicas.

Each of the three independently trained networks exceeds the energy coordinate on its own (63%, 67%, 70% versus 40% for a single 30-replica WE[E] draw), and pooling the three networks gives 60/90 (66.7%). That 30-replica WE[E] figure is a single sample, though, reused as the comparison point for all three networks, so an expanded, independent WE[E] baseline of 90 replicas is used for the balanced comparison reported here: WE[E] reaches 44/90 (48.9%) at the larger sample size. Fisher’s exact test on the balanced comparison gives

$$\text{WE}[I_\theta] : 60/90 (66.7\%) \text{ versus } \text{WE}[E] : 44/90 (48.9\%), \quad p = 0.023. \quad (12)$$

This is the central within-disorder result: the learned importance map significantly increases the probability that a finite-budget WE run reaches the deep target, reproducibly across three independent initializations.

The comparison is informative because energy is a strong physical baseline for this task: the target itself is defined by low energy, so an improvement over WE[E] is not a comparison against an irrelevant coordinate. The network receives the full spin configuration and can distinguish configurations with similar instantaneous energy but different local structure and different dynamical prospects of progressing toward the target. Two configurations at identical energy can have very different committor values on a frustrated landscape, after all, and an energy-adjacent label cannot represent that difference. The role of the learned function remains conservative throughout: it defines the WE partition rather than replacing the physical transition kernel, so the estimator remains unbiased for any binning coordinate, and the learned model influences only how computational effort is distributed across configuration space.

5.2. Generalization across disorder realizations

The result above establishes reproducibility across network initializations on one J_{ij} sample; it does not by itself establish that the effect is a property of the method rather than of that particular disorder realization. Five further independent EA disorder realizations were tested at the same $L = 16$, $T = 2.6$, beyond= 4 configuration for this reason, one trained network each. The network-initialization axis has already been shown not to be the dominant source of variance, so a single network per realization is the appropriate use of computational effort here rather than a further three-network ensemble per realization.

Disorder realization	WE[I_θ]	WE[E]	Fisher p
12345 (original)	19/30 (63.3%)	12/30 (40.0%)	0.120
23456	15/30 (50.0%)	8/30 (26.7%)	0.110
45678	29/30 (96.7%)	26/30 (86.7%)	0.353
56789	20/30 (66.7%)	15/30 (50.0%)	0.295
67890	28/30 (93.3%)	29/30 (96.7%)	1.000
78901	10/30 (33.3%)	5/30 (16.7%)	0.233
Pooled (6 realizations)	121/180 (67.2%)	95/180 (52.8%)	0.0071

Table 5: Deep-rare EA comparison repeated across six independent disorder realizations, one trained network each, beyond= 4, 30 replicas per method. Realization 12345 is the single-network reading of Table 4 (seed 0), not the three-network pooled result.

Deep-rare EA discovery rate ($L = 16$, beyond = 4): error bars are Wilson 95% intervals; diamonds in (a) are the 3 individual trained networks

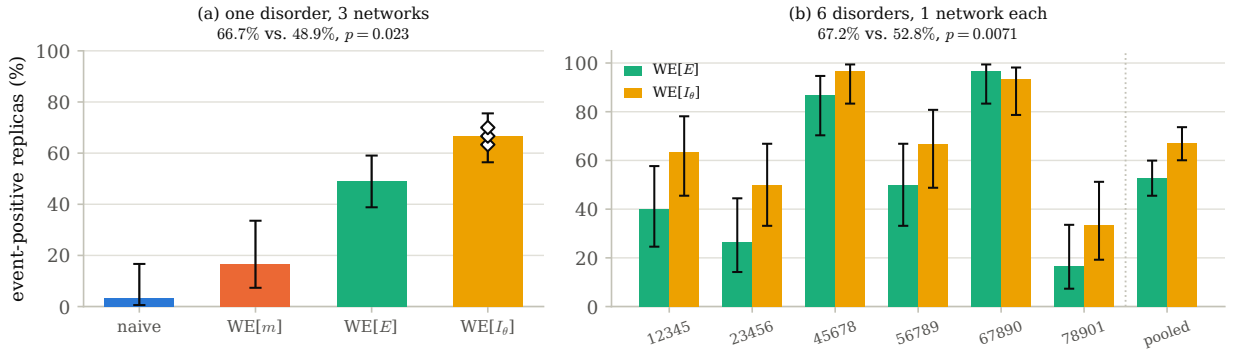


Figure 1: The learned importance coordinate reaches the deep target more reliably than the physical energy coordinate. Error bars are Wilson 95% intervals. (a) Discovery rate at beyond = 4 (disorder realization 12345): open diamonds are the three independently trained networks (Table 4); WE[E] uses the balanced 90-replica baseline. (b) The same comparison repeated across six independent disorder realizations, one trained network each (Table 5), with the pooled result at right.

No single realization reaches significance on its own at $n = 30$: the original realization, read as a single-network comparison rather than the three-network pooled result of §5.1, is itself not significant ($p = 0.120$). That is the expected pattern at this replica count, not a weakening of the result. Realizations 45678 and 67890 present a substantially easier calibrated target for their particular disorder, with both methods near ceiling in each case, which is why their own tests are the least informative of the six; this is a ceiling effect, not evidence against the finding, and it is also the one place the pattern reverses: realization 67890 nominally favors WE[E] (96.7% versus 93.3%), though at that discovery rate the comparison carries little information either way. In the other five of six realizations the learned coordinate has the higher discovery rate. Pooled across all six independent realizations, one network each (180 replicas per method):

$$\text{WE}[I_{\theta}] : 121/180 (67.2\%) \text{ versus } \text{WE}[E] : 95/180 (52.8\%), \quad p = 0.0071. \quad (13)$$

This is a different and stronger pooling axis than §5.1. Rather than asking whether the effect is robust to network initialization on one lattice, it asks whether the effect is a property of the method across the disorder ensemble. Wall-clock cost ratios for the five new realizations (4.45–4.97 relative to WE[E]) are consistent with every other measurement in this study (§6.4), so the efficiency conclusion generalizes together with the reliability one. The result should be read as reproduced across the six independent disorder realizations tested, not as a universal property of the EA spin glass at this scale.

One candidate disorder realization was set aside during this study because its collection stage produced a training set with no bulk-reference states activating the second boundary condition of §2.5, so the self-consistency objective could not rule out the trivial constant solution. This was traced to a fixable limitation in how the training set was subsampled, not to a property of that disorder realization: whenever the number of boundary-adjacent states collected exceeded the training-set size, all of it was filled with boundary states and none with bulk states, regardless of how many bulk states were available in the wider collection pool. Capping boundary-state inclusion at a fixed fraction of the training-set size resolves this, and every realization reported here trained without incident once that cap was in place.

6. Discussion

6.1. What is established

Five elements of the transport-inspired framework are established by direct test. Exact importance gives the zero-variance first-passage ideal (§3.1). Weight-conserving WE is unbiased in the exact enumeration and learned-coordinate gates (§3.2). WE holds a real, roughly two-fold precision advantage over naive sampling at the deepest resolved point

of the Ising tail at matched update cost (§4.1). The learned importance coordinate improves deep-target discovery relative to energy in the EA spin glass (§5.1). That reliability advantage is reproduced across six independent disorder realizations, not tied to network initialization or to one quenched-disorder sample (§5.2).

6.2. What is not established

The learned coordinate is not wall-clock superior (§6.4). WE’s precision advantage over naive sampling in the Ising tail is confined to the single deepest resolved bin, not a uniform improvement across the tail (§4.1); an earlier characterization at fewer replicas overstated this as a systematic advantage, which does not survive replication. Only one lattice size, one temperature, and a modest ($n = 6$) disorder ensemble have been studied for the main learned-coordinate result, and that ensemble is not large enough to characterize how the effect’s magnitude varies with disorder. The learned coordinate has not been demonstrated to be quantitatively close to the true committor; only that its use as a binning coordinate improves access, which does not require that accuracy (§2.5).

6.3. Why this matters

A coordinate can be dynamically more informative, in the sense of organizing a weighted population so that it reaches a target more often, while being computationally less efficient once every real cost is counted. Reliability and efficiency are not the same axis. Conflating them would have overstated the practical readiness of the learned coordinate at this scale while understating its scientific content: on a frustrated, quenched-disordered landscape where the scalar energy baseline is known to be weak, it demonstrably extracts progress information beyond that baseline.

6.4. Reliability is improved; cost-normalized efficiency is not

On the spin-update cost measure, the three within-disorder trained networks give figure-of-merit ratios of 1.29, 1.07, and 1.04 against $\text{WE}[E]$, bootstrapping to a geometric-mean ratio of 1.16 (95% CI [0.65, 2.12]) that does not exclude parity. That measure does not charge for network evaluations performed at every resampling step, though, so wall-clock time was recorded directly (not reconstructed from log timestamps) for every method. Measured ratios of $\text{WE}[I_\theta]$ to $\text{WE}[E]$ wall-clock time were 4.65, 4.01, and 4.54 for the three within-disorder networks, and 4.45, 4.97, 4.93, 4.80, and 4.48 for the five additional disorder realizations: consistently four to five times $\text{WE}[E]$ ’s cost. A two-level bootstrap gives the wall-clock-fair figure below, resampling the native 30-replica count from the balanced 90-replica $\text{WE}[E]$ pool at its own measured wall-clock cost, so both sides of the ratio are charged like-for-like:

$$\text{FOM}_{\text{wall}}(\text{WE}[I_\theta]) / \text{FOM}_{\text{wall}}(\text{WE}[E]) = 0.33, \quad 95\% \text{ CI} = [0.19, 0.58], \quad (14)$$

below parity and excluding 1. A separate, more heavily instrumented five-network ensemble run at a shallower configuration showed the same effect more strongly still (wall-clock ratio 10.7–10.9 $\times$, FOM_{wall} ratio ≈ 0.087). Additional network capacity moves the cost penalty in the wrong direction faster than it can improve reliability, so that number is reported here only as a scaling indication, not as a second deep-rare measurement. A learned coordinate can be dynamically better and simultaneously worse per unit of computer time, in other words. Reducing inference cost, whether through caching, batching, or a smaller network, rather than improving the coordinate further, is the binding constraint on making learned importance practical at this scale, and is the natural target of a dedicated follow-up efficiency study with timing recorded as a first-class output throughout.

6.5. Relation to neighboring rare-event methods

The present implementation is one member of a broader splitting and path-sampling family. FFS and AMS advance trajectories through fixed interfaces or adaptively chosen levels [20–22], and transition-path sampling works directly with reactive-path ensembles, which frees it from a fixed scalar coordinate but targets different estimators than the stationary tail probabilities studied here [1, 23, 45]. WE differs in maintaining a continuously weighted population and resampling within bins, so that the bins influence variance but not the target measure [16, 17, 38]. What the transport analogy adds is a principled target for choosing those bins: CADIS couples an importance map to source biasing and weight windows [10, 11, 15], and the present implementation adopts the population-control half of that logic without altering the Metropolis kernel.

Against the neural rare-event literature, the distinction is where the learned object is inserted. Flow-based and autoregressive samplers learn global equilibrium structure [24–27]; neural committor and Feynman–Kac methods target dynamical progress toward a specific event [28, 29, 39]; and active or self-consistent committor schemes concentrate training and sampling effort in transition-relevant regions [40–42]. The framework of Kim and Cai is

closest in spirit, combining a learned event-specific importance function with biased transitions and branching [31]: in the language of §2.1, their importance function plays the role of h , their modified transitions the role of the tilt, and their branching the role of splitting. All of these place the learned function inside the dynamics. The present approach places it only in the binning instead, which preserves unbiasedness through weight conservation and makes coordinate quality measurable without confounding it with likelihood-ratio error.

6.6. Scope and reproducibility

The principal learned-coordinate experiment uses six two-dimensional EA disorder realizations, one temperature ($T = 2.6$), one lattice size ($L = 16$), one threshold depth (beyond= 4), and a compact periodic convolutional network. Replication is across trained networks, disorder realizations, and simulation replicas, not yet across lattice size or temperature. A larger disorder sample, to tighten the pooled confidence interval and better characterize how the effect's magnitude varies with disorder, and extensions in L and T , are the natural next steps. Because outcomes vary appreciably between training initializations at this scale, a single training run is not evidence about a learned coordinate in either direction; seed-resolved reporting is treated as a requirement throughout this study, and the training-set fix of §5.2 is recommended practice for the same reason when extending this result further.

7. Conclusions

Adjoint transport theory provides a useful organizing principle for rare-event sampling in lattice spin systems. The committor supplies the event-specific importance function, the exact Doob transform defines the zero-variance ideal, and weight-conserving population control provides a practical way to exploit approximate importance without changing the target measure; these elements were validated sequentially from a solvable birth–death process to exact enumeration of the Ising model.

At system scale, WE and naive sampling resolve the $L = 20$ Ising magnetization tail to comparable depth once properly replicated ($R = 20$): WE holds a real but modest, roughly two-fold precision advantage at the single deepest bin either method reaches, not a uniform advantage across the tail. An earlier characterization at fewer replicas reported this as a systematic $\sim 24\times$ extension; that framing does not survive replication and is retracted here.

In the frustrated EA model, the learned configuration-space importance coordinate produces the main new result of this study. Pooled across six independent disorder realizations with one trained network each, it reaches the deep target in 121/180 replicas against 95/180 for the physical energy coordinate (67.2% versus 52.8%, Fisher exact $p = 0.0071$), reproducing a within-disorder result across three independently trained networks against a balanced baseline (66.7% versus 48.9%, $p = 0.023$). A learned progress coordinate can extract dynamical information not fully represented by a scalar physical observable, then, and this advantage is a property of the method rather than of one particular quenched-disorder sample.

That advantage is presently one of reliability rather than efficiency: repeated network inference is directly measured to cost four to five times WE[E]'s wall-clock time, leaving the learned coordinate at roughly one-third of parity in figure of merit (0.33, 95% CI [0.19, 0.58]). Reducing that inference cost, rather than improving the coordinate further, is the binding constraint on making learned importance practical at this scale.

Data and Code Availability

All code, input data, and result files required to reproduce the results reported here are publicly archived [46]. The deposit contains the Weighted Ensemble implementation and production drivers, the training code, the quenched disorder realizations used for every Edwards–Anderson result, per-replica $\hat{\pi}$ arrays for every table and figure, the exact-enumeration and birth–death validation outputs, the trained network weights for the three within-disorder networks and the five additional disorder-realization networks of §5.2, and the raw wall-clock logs underlying §6.4. An environment specification pins package versions, and all published results are CPU-deterministic under the recorded seeds.

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CRediT authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

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